Bayesian Framework for Canonical factorization of multiblock data

1 Introduction

Data sets structured into blocks of variables measured on the same set of observations are frequently encountered in practice. These blocks may consist of quantitative variables originating from distinct measurement sources or correspond to thematic groupings defined by the user. For instance, in omics studies, blocks typically comprise variables derived from different analytical techniques (e.g., proteomics, metabolomics). In psychology, multiblock data commonly arise when several sets of variables are collected on the same individuals to capture different psychological domains, such as self-report questionnaire scores assessing personality traits, behavioral task performance measures, and neurophysiological indicators such as electroencephalography features.

Exploratory multiblock data analysis has four main objectives: (i) to perform a joint dimensionality reduction, (ii) to examine relationships among blocks, (iii) to investigate associations among variables both within and across blocks, and (iv) to capture within-block variability in order to reveal shared and block-specific structures in a common latent space. A unified algebraic framework based on factorization strategies, called *canonical factorization*, was proposed by [Bougeard et al., 2024] for multiblock methods to achieve these objectives. In practice, canonical factorization provides a structured framework for analyzing data variability at multiple levels: differences between and within blocks, and differences between observations through their projection onto a common subspace. It also facilitates the harmonization of the outputs of different methods and supports a criterion-based selection among competing approaches.

The canonical factorization is exact when the dimension of the common subspace equals the rank of the concatenated matrix, which can be large. Determining a reduced dimension, and identifying which dimensions are common to all blocks, shared by some of them or specific to one block, is therefore a major issue. Bayesian approaches address the first question for a single matrix: [Hoff, 2007] proposed a Bayesian singular value decomposition (SVD) that quantifies the uncertainty on the rank, and competing approaches were obtained with other priors on the singular values, such as the double exponential distribution [Zhai and Gutman, 2023], the repulsed normal distribution [Yuchi et al., 2023] and the horseshoe prior [Wang et al., 2026], which shrinks small singular values towards zero.

Following this line of work, we propose a Bayesian framework for the canonical factorization of multiblock data. The model imposes orthonormality constraints on both the common components and the block loadings, which makes the axes of the common subspace identifiable and leads to matrix von Mises–Fisher full conditional distributions. We derive an efficient Gibbs sampler and show that its conditional modes coincide with the updates of Optimal Multiple Co-Inertia Analysis (OMCOA), the simultaneous alternative to Multiple Co-Inertia Analysis [Chessel and Hanafi, 1996]. We then propose a sparse version based on a spike-and-slab prior [Mitchell and Beauchamp, 1988, George and McCulloch, 1993] on the block weights. Unlike continuous shrinkage priors, this prior yields exact zeros and posterior inclusion probabilities, which directly reveal common, partially common and specific structures and provide an estimate of the dimension of the common subspace.

2 Canonical factorisation

Notations

- K blocks $(\mathbf{X}_1, \ldots, \mathbf{X}_K)$ respectively of $p_1, \ldots, p_K$ variables on N individuals,
- $\mathbf{X} = [\mathbf{X}_1 | \ldots | \mathbf{X}_K] \in \mathcal{M}_{N,P}(\mathbb{R})$ $(P = \sum_{k=1}^{K} p_k)$ be the concatenation of these K-blocks,
- $\mathbf{U}^\intercal$ designates the transposed matrix of $\mathbf{U}$,
- $\Sigma_k = diag(\sigma_k^{(1)}, ..., \sigma_k^{(R)})$ are diagonal matrices
- $\|.\|$ correspond to the Frobenius norm,
- $\mathcal{V}_{N,R} = \{\mathbf{W} \in \mathcal{M}_{N,R}(\mathbb{R}) : \mathbf{W}^\intercal \mathbf{W} = \mathbf{I}_R\}$ denotes the Stiefel manifold.

Consider K colmumn blocks $(\mathbf{X}_1, \ldots, \mathbf{X}_K)$ respectively in $\mathcal{M}_{N,p_1}(\mathbb{R}), ..., \mathcal{M}_{N,p_K}(\mathbb{R})$. Let $\mathbf{X} = [\mathbf{X}_1 | \ldots | \mathbf{X}_K] \in \mathcal{M}_{N,P}(\mathbb{R})$ $(P = \sum_{k=1}^{K} p_k)$ be the concatenation of these K-blocks. Without loss of generality, the blocks are assumed to be column-centered.

[Bougeard et al., 2024] demonstrated that the $(K+1)-$blocks $(\mathbf{X}_1, \ldots, \mathbf{X}_K, \mathbf{X})$ can be factorized in dimension $R = rank(\mathbf{X})$ as follows

$$\begin{cases} \mathbf{X} = \mathbf{U}\Sigma\mathbf{V}^\intercal \\ \mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal \quad \text{for} \quad k = (1, \ldots, K) \\ \text{with the constraints} \quad \mathbf{U} \in \mathcal{V}_{N,R} \quad \text{and} \quad \|\mathbf{v}_k^{(1)}\| = \ldots = \|\mathbf{v}_k^{(R)}\| = 1 \quad \text{for} \quad k = (1, \ldots, K), \end{cases} \tag{1}$$

where

- $\mathbf{U} \in \mathcal{V}_{N,R}$ is the block-common components,
- $\mathbf{V}_k \in \mathcal{M}_{p_k,R}(\mathbb{R})$ are the block-loadings, and $\mathbf{V} = [\mathbf{V}_1 | ... | \mathbf{V}_K] \in \mathcal{M}_{P,R}(\mathbb{R})$,
- $\Sigma \in \mathcal{M}_{R,R}(\mathbb{R})$ the diagonal scaling matrices with property $\Sigma^2 = \sum_{k=1}^{K} \Sigma_k^2$ that relate common and block-analyses.

Without loss of generality, we can assume $\sigma_k^{(r)} \geq 0$ for all k, r, since a joint sign change of $\sigma_k^{(r)}$ and $\mathbf{v}_k^{(r)}$ leaves the model and the constraints unchanged. Since the blocks $\mathbf{X}_1, ...\mathbf{X}_K$ are column-centered, the block-common $\mathbf{U} \in \mathcal{V}_{N,R}$ is centered as-well:

$$\mathbf{U} \in \mathcal{V}_{N,R}, \quad \mathbf{1}_N\mathbf{U} = 0.$$

This factorization is *the canonical factorization*. Each multiblock method derives a common component $\mathbf{U}$ using a different optimization criterion. Projecting the data onto these common components enables their canonical factorization (for details see [Bougeard et al., 2024]). For all the multiblock methods, once $\mathbf{U}$ is fixed, the decomposition $\mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal$ of each block is unique (up to sign). The methods differ in how the axes $\mathbf{u}^{(1)}, \ldots, \mathbf{u}^{(R)}$ themselves are determined. When the block loadings are only normalized ($\|\mathbf{v}_k^{(r)}\| = 1$), the optimal loadings are $\mathbf{v}_k^{(r)} = \mathbf{X}_k^\intercal\mathbf{u}^{(r)} / \|\mathbf{X}_k^\intercal\mathbf{u}^{(r)}\|$ and the least-squares criterion reduces to

$$\sum_{k=1}^{K} \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X}\|^2 - \|\mathbf{X}^\intercal\mathbf{U}\|^2,$$

which is invariant under any rotation $\mathbf{U} \mapsto \mathbf{UR}$: only span($\mathbf{U}$) is determined by the data, and the axes have to be fixed by an external convention, usually the SVD of $\mathbf{X}$. This convention becomes unstable when the singular values of $\mathbf{X}$ are close. When $\mathbf{V}_k \in \mathcal{V}_{p_k,R}$ for all k, the criterion is no longer rotation invariant, and the axes of $\mathbf{U}$ are identified (up to permutation and sign) by the block-specific profiles $\Sigma_1, \ldots, \Sigma_K$, as long as these profiles differ across blocks.

2.1 Canonical Factorisation under orthogonal block-loadings constraint

Providing a solution for canonical factorization means minimizing

$$\sum_{k=1}^{K} \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X} - \mathbf{U}\Sigma\mathbf{V}^\intercal\|^2, \tag{2}$$

with $\mathbf{U} \in \mathcal{V}_{N,R}$, $\mathbf{V}_k \in \mathcal{V}_{p_k,N}$ for $k = 1, \ldots, K$. Chessel and Hanafi [Chessel and Hanafi, 1996] proposed a sequential solution, known as Multiple Co-Inertia Analysis (MCOA): the dimensions are computed one at a time, each from the blocks deflated with respect to the previous block loadings. This procedure yields orthonormal matrices $\mathbf{U}$ and $\mathbf{V}_1, \ldots, \mathbf{V}_K$, and each stage is optimal in the least-squares sense given the previous ones. The resulting solution is nevertheless suboptimal with respect to (2), since each dimension is chosen without taking the next ones into account, so that the R dimensions are not fitted simultaneously. Optimal Multiple COinertia Ananlysis (OMCOA) is proposed as a simultaneous solution of (2) under the constraints $\mathbf{U} \in \mathcal{V}_{N,R}$ and $\mathbf{V}_k \in \mathcal{V}_{N,R}$ ($k = 1, \ldots, K$). It is computed by an alternating least squares (ALS) algorithm, in which each step minimizes (2) with respect to one set of parameters, the others being fixed:

- **Update of U:** $\mathbf{U} = \mathbf{AB}^\intercal$, where $\mathbf{XH}^\intercal = \mathbf{ADB}^\intercal$ is the SVD of $\mathbf{XH}^\intercal$, with $\mathbf{H} = [\Sigma_1\mathbf{V}_1^\intercal | \cdots | \Sigma_K\mathbf{V}_K^\intercal]$ (orthogonal Procrustes problem);
- **Update of $\mathbf{V}_k$:** $\mathbf{V}_k = \mathbf{A}_k\mathbf{B}_k^\intercal$, where $\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k = \mathbf{A}_k\mathbf{D}_k\mathbf{B}_k^\intercal$ is the SVD of $\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k$, for $k = 1, \ldots, K$;
- **Update of Σ_k:** $\Sigma_k = \mathrm{diag}(\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)$, for $k = 1, \ldots, K$.

Proposition 1. *The successive updates of $\mathbf{U}, \mathbf{V}_k, \Sigma_k$ guarantee that the criterion defined in* (2) *decreases or remains constant at each step, ensuring convergence of the algorithm.*

The proof is given in the appendix.

Remark 1. *Since the limit may however be a local minimum, the algorithm is started from the MCOA solution and from several random initializations, and the solution with the lowest criterion is retained.*

3 Bayesian framework for Canonical Factorisation

A probabilistic counterpart of (2) is obtained by assuming a Gaussian noise:

$$\mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal + \mathbf{E}_k, \qquad \mathbf{E}_k \sim \mathcal{N}(0, \tau_k^{-1}), \qquad k = 1, \ldots, K, \tag{3}$$

where τ_k is the noise precision of block k. centered as well: $\mathbf{U} \in \mathcal{V}_{N,R}$ with $\mathbf{1}_N^\intercal\mathbf{U} = 0$.

Priors. We use the following independent priors:

- $\mathbf{U}$ and $\mathbf{V}_1, \dots, \mathbf{V}_K$ are uniformly distributed on their Stiefel manifolds (non-informative von Mises–Fisher distribution). A random matrix $\mathbf{W} \in \mathcal{V}_{m,R}$ follows the matrix von Mises–Fisher distribution $\mathrm{vMF}(\mathbf{F})$, with parameter $\mathbf{F} \in \mathcal{M}_{m,R}(\mathbb{R})$, if its density with respect to the uniform measure on $\mathcal{V}_{m,R}$ is proportional to $\exp\{\mathrm{trace}(\mathbf{F}^\intercal \mathbf{W})\}$ [Hoff, 2009]. If $\mathbf{F} = \mathbf{A}\mathbf{D}\mathbf{B}^\intercal$ is the SVD of $\mathbf{F}$, the mode of $\mathrm{vMF}(\mathbf{F})$ is the polar factor $\mathbf{A}\mathbf{B}^\intercal$, and the distribution concentrates around it as the singular values of $\mathbf{F}$ increase.
- $\sigma_k^{(r)} \sim \mathcal{N}^+(0, \lambda^{-1})$, a half-normal distribution with a small precision λ (e.g. $\lambda = 10^{-2}$);
- $\tau_k \sim \mathrm{Gamma}(a_0, b_0)$ with small hyperparameters (e.g. $a_0 = b_0 = 10^{-3}$).

3.1 Full conditional distributions

Likelihood. Because $\mathbf{U}$ and $\mathbf{V}_k$ have orthonormal columns and Σ_k is diagonal,

$$\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X}_k\|^2 - 2\,\mathrm{trace}(\Sigma_k\mathbf{V}_k^\intercal\mathbf{X}_k^\intercal\mathbf{U}) + \sum_{r=1}^{R}(\sigma_k^{(r)})^2. \tag{4}$$

Since the blocks are centered, each $\mathbf{X}_k$ carries $(N-1)p_k$ degrees of freedom, and the log-likelihood is

$$\log p(\mathbf{X}_1, \dots, \mathbf{X}_K \mid \cdot) = \sum_{k=1}^{K}\Big[\tfrac{(N-1)p_k}{2}\log\tau_k - \tfrac{\tau_k}{2}\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2\Big] + \text{const.}$$

By (4), $\mathbf{U}$ and $\mathbf{V}_k$ only appear in the linear term $\mathrm{trace}(\Sigma_k\mathbf{V}_k^\intercal\mathbf{X}_k^\intercal\mathbf{U})$. This is the key property: all the full conditionals belong to standard families.

Proposition 2 (Full conditionals)**.** *Under model* (3) *and the above priors:*

1. ***Common space.*** *Let $\mathbf{Q} \in \mathcal{M}_{N,N-1}(\mathbb{R})$ be an orthonormal basis of $\mathbf{1}_N^\perp$ and write $\mathbf{U} = \mathbf{Q}\tilde{\mathbf{U}}$ with $\tilde{\mathbf{U}} \in \mathcal{V}_{N-1,R}$. Then*

$$\tilde{\mathbf{U}} \mid \cdot \;\sim\; \mathrm{vMF}(\mathbf{Q}^\intercal\mathbf{F}_{\mathbf{U}}), \qquad \mathbf{F}_{\mathbf{U}} = \sum_{k=1}^{K}\tau_k\,\mathbf{X}_k\mathbf{V}_k\Sigma_k.$$

2. ***Block loadings.*** *For $k = 1, \dots, K$,*

$$\mathbf{V}_k \mid \cdot \;\sim\; \mathrm{vMF}(\mathbf{F}_{\mathbf{V}_k}), \qquad \mathbf{F}_{\mathbf{V}_k} = \tau_k\,\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k.$$

3. ***Block weights.*** *For $k = 1, \dots, K$ and $h = 1, \dots, H$, let $d_k^{(r)} = \mathbf{u}^{(r)\intercal}\mathbf{X}_k\mathbf{v}_k^{(r)}$. Then*

$$\sigma_k^{(r)} \mid \cdot \;\sim\; \mathcal{N}^+\Big(\frac{\tau_k\,d_k^{(r)}}{\tau_k + \lambda},\; \frac{1}{\tau_k + \lambda}\Big),$$

   a normal distribution truncated to $(0, +\infty)$.

4. ***Noise precisions.*** *For $k = 1, \dots, K$,*

$$\tau_k \mid \cdot \;\sim\; \mathrm{Gamma}\Big(a_0 + \frac{(N-1)p_k}{2},\; b_0 + \frac{1}{2}\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2\Big).$$

The modes of the full conditionals of Proposition 2 are

$$\mathbf{U} = \mathbf{A}\mathbf{B}^\intercal \text{ with } \mathbf{F}_\mathbf{U} = \mathbf{A}\mathbf{D}\mathbf{B}^\intercal, \qquad \mathbf{V}_k = \mathbf{A}_k\mathbf{B}_k^\intercal \text{ with } \mathbf{F}_{\mathbf{V}_k} = \mathbf{A}_k\mathbf{D}_k\mathbf{B}_k^\intercal, \qquad \sigma_k^{(h)} = \frac{\tau_k}{\tau_k + \lambda}\, d_k^{(h)}.$$

When $\lambda \to 0$ and the precisions are equal ($\tau_1 = \cdots = \tau_K$), these are exactly the OMCOA updates: Procrustes rotation of $\mathbf{X}\mathbf{H}^\intercal$ for $\mathbf{U}$ (with $\mathbf{H} = [\Sigma_1\mathbf{V}_1^\intercal | \cdots | \Sigma_K\mathbf{V}_K^\intercal]$), Procrustes rotation of $\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k$ for $\mathbf{V}_k$, and $\Sigma_k = \mathrm{diag}(\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)$. OMCOA is therefore the iterated conditional modes (ICM) algorithm [Besag, 1986] associated with this posterior, and the Gibbs sampler is its stochastic counterpart. With block-specific precisions, $\mathbf{F}_\mathbf{U} = \sum_k \tau_k\mathbf{X}_k\mathbf{V}_k\Sigma_k$ weights each block by its estimated noise precision, so that noisier blocks contribute less to the common space. To reproduce exactly the OMCOA criterion (2), a common precision $\tau_1 = \cdots = \tau_K = \tau$ can be used, with (2), $\tau \mid \cdot \sim \mathrm{Gamma}\big(a_0 + \sum_k (N-1)p_k/2,\ b_0 + \sum_k \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2/2\big)$.

Post-processing: label switching and signs The posterior distribution is invariant under any permutation π of the R dimensions applied jointly to $\mathbf{U}$, $\mathbf{V}_1, \ldots, \mathbf{V}_K$ and $\Sigma_1, \ldots, \Sigma_K$, and under a joint change of sign of $\mathbf{u}^{(r)}$ and $\mathbf{v}_1^{(r)}, \ldots, \mathbf{v}_K^{(r)}$. Posterior summaries are therefore computed after aligning each draw on a reference $\mathbf{U}_{\mathrm{ref}}$ (in practice the OMCOA solution). For each draw, with $\mathbf{C} = \mathbf{U}_{\mathrm{ref}}^\intercal\mathbf{U}$,

$$\hat{\pi} = \arg\max_{\pi \in \mathfrak{S}_R} \sum_{h=1}^{R} \big|C_{r,\pi(r)}\big|, \qquad s_r = \mathrm{sign}\big(C_{r,\hat{\pi}(r)}\big),$$

and the columns of $\mathbf{U}$ and $\mathbf{V}_k$ are reordered by $\hat{\pi}$ and multiplied by s_r, while the diagonal of Σ_k is reordered by $\hat{\pi}$ (it stays positive). Posterior means and credible intervals of $\sigma_k^{(r)}$ are then obtained from the empirical distribution of the aligned draws.

We can deduce the following algorithm:

Algorithm 1 Gibbs sampler for OMCOA

Require: centered blocks $\mathbf{X}_1, \ldots, \mathbf{X}_K$; dimension H; number of iterations T; burn-in T_0; hyperparameters a_0, b_0, λ

Initialization: $\mathbf{U} = \mathbf{Q}\tilde{\mathbf{U}}$, $\mathbf{V}_k$, Σ_k from OMCOA (or MCOA, or at random); $\tau_k \leftarrow (N-1)p_k / \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2$
for $t = 1, \ldots, T$ **do**
  $\mathbf{F}_\mathbf{U} \leftarrow \sum_k \tau_k\, \mathbf{X}_k\mathbf{V}_k\Sigma_k$; update $\tilde{\mathbf{U}}$ by one column-wise sweep of vMF($\mathbf{Q}^\intercal\mathbf{F}_\mathbf{U}$); $\mathbf{U} \leftarrow \mathbf{Q}\tilde{\mathbf{U}}$
  for $k = 1, \ldots, K$ **do**
    $\mathbf{F}_{\mathbf{V}_k} \leftarrow \tau_k\, \mathbf{X}_k^\intercal\mathbf{U}\Sigma_k$; update $\mathbf{V}_k$ by one column-wise sweep of vMF($\mathbf{F}_{\mathbf{V}_k}$)
    for $h = 1, \ldots, H$ **do**
      $d_k^{(h)} \leftarrow \mathbf{u}^{(h)\intercal}\mathbf{X}_k\mathbf{v}_k^{(h)}$; draw $\sigma_k^{(h)} \sim \mathcal{N}^+\big(\tau_k d_k^{(h)}/(\tau_k+\lambda),\ 1/(\tau_k+\lambda)\big)$
    end for
    draw $\tau_k \sim \mathrm{Gamma}\big(a_0 + (N-1)p_k/2,\ b_0 + \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2/2\big)$
  end for
  if $t > T_0$ **then** store $\big(\mathbf{U}, \mathbf{V}_1, \ldots, \mathbf{V}_K, \Sigma_1, \ldots, \Sigma_K, \tau_1, \ldots, \tau_K\big)$
  end if
end for
Post-processing: align each stored draw on a reference solution (Section 3.1)
return the aligned draws

4 Sparse Bayesian Framework for canonical factorisation

In OMCOA, each block is approximated as $\mathbf{X}_k \approx \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal$, where the diagonal matrix $\Sigma_k = \mathrm{diag}(\sigma_k^{(1)}, \dots, \sigma_k^{(R)})$ measures how much each latent dimension $\mathbf{u}^{(r)}$ contributes to block k. The structure of the blocks can be read from the pattern of zeros of $\Sigma_1, \dots, \Sigma_K$: dimension r is

- *U-common* if $\sigma_k^{(r)} > 0$ for all k;
- *partially U-common* if $\sigma_k^{(r)} > 0$ for some blocks only;
- *U-specific* to block k if $\sigma_k^{(r)} > 0$ for this block only;
- *inactive* if $\sigma_k^{(r)} = 0$ for all k.

OMCOA always returns strictly positive weights, so that this pattern has to be assessed afterwards. We propose a Bayesian version of OMCOA in which a spike-and-slab prior on the weights $\sigma_k^{(r)}$ allows them to be exactly zero. The dimension R is then a *maximal* rank: inactive dimensions are switched off by the sampler, and the effective rank is inferred from the data.

The sparsity of the model is based on the diagonal terms $\sigma_k^{(r)}$ as follows

$$\sigma_k^{(r)} = \gamma_k^{(r)}\, s_k^{(r)}, \qquad \gamma_k^{(r)} \in \{0,1\}, \quad s_k^{(r)} > 0. \tag{5}$$

Priors. The difference between the two sets of prior distributions lies in the prior distributions of the diagonal terms:

- spike-and-slab on the weights: $\gamma_k^{(r)} \mid \theta_r \sim \mathrm{Bernoulli}(\theta_r)$ and $s_k^{(r)} \sim \mathcal{N}^+(0, \lambda^{-1})$, i.e.

$$p(\sigma_k^{(r)} \mid \theta_r) = (1-\theta_r)\,\delta_0(\sigma_k^{(r)}) + \theta_r\,\mathcal{N}^+(\sigma_k^{(r)}; 0, \lambda^{-1});$$

- $\theta_r \sim \mathrm{Beta}(a_\theta, b_\theta)$, one inclusion probability per dimension, shared by the K blocks.

Sharing θ_r across blocks borrows strength: a dimension that is active in several blocks is more likely to be active in another one, and a dimension that is inactive everywhere sees θ_r shrink towards zero.

As in the non-sparse case, the orthonormality of $\mathbf{U}$ and $\mathbf{V}_k$ gives

$$\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X}_k\|^2 - 2\sum_{r=1}^{R}\sigma_k^{(r)}d_k^{(r)} + \sum_{r=1}^{R}(\sigma_k^{(r)})^2, \qquad d_k^{(r)} = \mathbf{u}^{(r)\intercal}\mathbf{X}_k\mathbf{v}_k^{(r)}. \tag{6}$$

Proposition 3 (Full conditionals)**.** *Let $\mathbf{Q} \in \mathcal{M}_{N,N-1}(\mathbb{R})$ be an orthonormal basis of $\mathbf{1}_N^\perp$ and write $\mathbf{U} = \mathbf{Q}\tilde{\mathbf{U}}$.*

1. ***Common space:*** $\tilde{\mathbf{U}} \mid \cdot \sim \mathrm{vMF}(\mathbf{Q}^\intercal\mathbf{F}_\mathbf{U})$ *with* $\mathbf{F}_\mathbf{U} = \sum_k \tau_k\mathbf{X}_k\mathbf{V}_k\Sigma_k$.
2. ***Block loadings:*** $\mathbf{V}_k \mid \cdot \sim \mathrm{vMF}(\mathbf{F}_{\mathbf{V}_k})$ *with* $\mathbf{F}_{\mathbf{V}_k} = \tau_k\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k$. *When $\sigma_k^{(r)} = 0$, the r-th column of $\mathbf{F}_{\mathbf{V}_k}$ is zero and $\mathbf{v}_k^{(r)}$ is uniform on the orthogonal complement of the other columns.*

3. ***Weights (joint update of $\gamma_k^{(r)}$ and $\sigma_k^{(r)}$).*** *Let*

$$\omega_k = \tau_k + \lambda, \qquad m_k^{(r)} = \frac{\tau_k \, d_k^{(r)}}{\omega_k}.$$

Then, with $\sigma_k^{(r)}$ integrated out,

$$P(\gamma_k^{(r)} = 1 \mid \cdot) = \frac{\theta_r B_k^{(r)}}{\theta_r B_k^{(r)} + 1 - \theta_r}, \qquad B_k^{(r)} = 2\sqrt{\frac{\lambda}{\omega_k}} \, \exp\Big(\frac{\omega_k (m_k^{(r)})^2}{2}\Big) \, \Phi\Big(m_k^{(r)} \sqrt{\omega_k}\Big),$$

where Φ is the standard normal c.d.f.; and

$$\sigma_k^{(r)} \mid \gamma_k^{(r)} = 1, \cdot \;\sim\; \mathcal{N}^+(m_k^{(r)}, \omega_k^{-1}), \qquad \sigma_k^{(r)} \mid \gamma_k^{(r)} = 0, \cdot \;=\; 0.$$

4. ***Inclusion probabilities:*** $\theta_r \mid \cdot \sim \mathrm{Beta}\big(a_\theta + \sum_k \gamma_k^{(r)},\; b_\theta + K - \sum_k \gamma_k^{(r)}\big)$.

5. ***Noise precisions:*** $\tau_k \mid \cdot \sim \mathrm{Gamma}\big(a_0 + (N-1)p_k/2,\; b_0 + \|\mathbf{X}_k - \mathbf{U}\Sigma_k \mathbf{V}_k^\intercal\|^2/2\big)$.

The matrix von Mises–Fisher distributions are sampled by one column-wise sweep per iteration, each column being drawn from a vector von Mises–Fisher distribution with Wood's algorithm, as in the non-sparse sampler. Updating $(\gamma_k^{(r)}, \sigma_k^{(r)})$ jointly, with $\sigma_k^{(r)}$ integrated out when drawing $\gamma_k^{(r)}$, is essential: drawing $\gamma_k^{(r)}$ given the current $\sigma_k^{(r)}$ would never allow the chain to leave $\sigma_k^{(r)} = 0$. In practice $\log B_k^{(r)}$ is computed on the log scale, with $\log \Phi$ evaluated directly, to avoid overflow.

From the posterior distributions of the parameter $\gamma_k^{(r)}$ a decision rule for the representation of the block $\mathbf{X}_k$ on the component $u^{(r)}$:

the component $\mathbf{u}^{(r)}$ is active for the block k if and only if $P(\gamma_k^{(r)} = 1 \mid \mathbf{X}_1, \ldots, \mathbf{X}_K) > 1/2$.

Moreover, this model makes it possible to estimate the rank of the blocks $(\mathbf{X}_1, ..., \mathbf{X}_K)$ i.e. $R_{\text{eff}} = \#\{r : \sum_k \gamma_k^{(r)} > 0\}$, whose posterior distribution is estimated by its empirical distribution over the draws.

We can deduce Sparse Gibbs sampler algorithm for OMCOA (algorithm 2).

5 Empirical performance via simulation

5.1 Simulation design

$K = 3$ blocks with $p_1 = 5$, $p_2 = 15$, $p_3 = 20$ variables and $N = 100$ observations are generated from

$$\mathbf{X}_k = \mathbf{U}\Sigma_k \mathbf{V}_k^\intercal + \mathbf{E}_k, \qquad k = 1, 2, 3,$$

where $\mathbf{U} \in \mathcal{V}_{N,3}$ is column-centered, $\mathbf{V}_k \in \mathcal{V}_{p_k,3}$, Σ_k is the diagonal matrix of the block weights and $\mathbf{E}_k$ is a Gaussian noise term; the true effective rank is $R_{\text{eff}} = 3$. Two cases are considered:

- case 1: $\Sigma_1 = \mathrm{diag}(0.7, 0.5, 0)$, $\Sigma_2 = \mathrm{diag}(0.6, 0.4, 0)$, $\Sigma_3 = \mathrm{diag}(0.5, 0, 0.6)$;

Algorithm 2 Sparse Gibbs sampler for OMCOA

Require: centered blocks $\mathbf{X}_1, \dots, \mathbf{X}_K$; maximal rank $R \le \min_k p_k$; iterations T; burn-in T_0; hyperparameters $a_0, b_0, \lambda, a_\theta, b_\theta$

Initialization: $\mathbf{U}, \mathbf{V}_k, \Sigma_k$ from OMCOA with R dimensions; $\gamma_k^{(r)} \leftarrow 1$; $\theta_r \leftarrow 1/2$; $\tau_k \leftarrow (N-1)p_k / \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2$
for $t = 1, \dots, T$ **do**
  update $\tilde{\mathbf{U}}$ by one sweep of $\mathrm{vMF}\big(\mathbf{Q}^\intercal \sum_k \tau_k \mathbf{X}_k \mathbf{V}_k \Sigma_k\big)$; $\mathbf{U} \leftarrow \mathbf{Q}\tilde{\mathbf{U}}$
  for $k = 1, \dots, K$ **do**
    update $\mathbf{V}_k$ by one sweep of $\mathrm{vMF}\big(\tau_k \mathbf{X}_k^\intercal \mathbf{U} \Sigma_k\big)$
    for $r = 1, \dots, R$ **do**
      $d_k^{(r)} \leftarrow \mathbf{u}^{(r)\intercal} \mathbf{X}_k \mathbf{v}_k^{(r)}$, $\omega_k \leftarrow \tau_k + \lambda$, $m_k^{(r)} \leftarrow \tau_k d_k^{(r)} / \omega_k$
      draw $\gamma_k^{(r)} \sim \mathrm{Bernoulli}\Big(\dfrac{\theta_r B_k^{(r)}}{\theta_r B_k^{(r)} + 1 - \theta_r}\Big)$
      if $\gamma_k^{(r)} = 1$ **then** draw $\sigma_k^{(r)} \sim \mathcal{N}^+\big(m_k^{(r)}, \omega_k^{-1}\big)$ **else** $\sigma_k^{(r)} \leftarrow 0$
    end for
    draw $\tau_k \sim \mathrm{Gamma}\big(a_0 + (N-1)p_k/2,\ b_0 + \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2/2\big)$
  end for
  for $r = 1, \dots, R$ **do**
    draw $\theta_r \sim \mathrm{Beta}\big(a_\theta + \sum_k \gamma_k^{(r)},\ b_\theta + K - \sum_k \gamma_k^{(r)}\big)$
  end for
  if $t > T_0$ **then** store $\big(\mathbf{U}, \Sigma_1, \dots, \Sigma_K, \boldsymbol{\gamma}, \boldsymbol{\theta}, \boldsymbol{\tau}\big)$
  end if
end for
align the stored draws (permutation and signs) on a reference $\mathbf{U}_{\mathrm{ref}}$
return the aligned draws

- case 2: $\Sigma_1 = \mathrm{diag}(0.7, 0.1, 0)$, $\Sigma_2 = \mathrm{diag}(0.6, 0.1, 0)$, $\Sigma_3 = \mathrm{diag}(0.5, 0, 0.1)$.

In both cases, the first dimension is common to the three blocks, the second one is shared by blocks 1 and 2, and the third one is specific to block 3. In case 2, the weights of the second and third dimensions are small: for instance, the third dimension explains about 3.5% of the variance of block 3, less than the noise. In all simulations, the noise is scaled so that $\|\mathbf{E}_k\|^2 = 0.1\,\|X_k\|^2$. The sparse model is fitted with a maximal rank $R_{\max} = 5$, i.e. the largest value allowed by $R_{\max} \le \min_k p_k$, with $T = 6000$ iterations, a burn-in of $T_0 = 1000$, $\lambda = 1$ and $a_\theta = b_\theta = 1$; the draws are aligned on the true $\mathbf{U}$ (permutation and signs of the dimensions) before computing posterior summaries. $B = 1000$ data sets are simulated in each case.

5.2 Results

Table 1 reports, for each weight, the mean over the B replications of the posterior mean $\hat{\sigma}_k^{(r)}$, its standard deviation across replications, and the percentage of replications in which dimension r is selected in block k, i.e. $\hat{P}(\gamma_k^{(r)} = 1 \mid \mathbf{X}_1, \dots, \mathbf{X}_K) > 1/2$.

The weights are accurately estimated in both cases, with no visible bias except a slight underestimation of the smallest weight (mean estimate 0.09 for a true value of 0.10 in case 2, dimension 3, block 3), as expected from a shrinkage prior. The selection rule has a very high empirical sen-

sitivity (proportion of active weights that are selected): 100% in case 1 and 99.1% in case 2. Its empirical specificity (proportion of null weights that are not selected) is also very high: 99.6% in case 1 and 99.1% in case 2. In the most difficult configuration (case 2, dimension 3, block 3), the specific dimension is selected in 94.9% of the replications; in most of the remaining ones, the signal is captured by the fourth dimension (4.7%), which suggests a labelling issue in the alignment rather than a failure of detection.

Figure 1 shows the distribution over the replications of the posterior probabilities $\hat{P}(R_{\text{eff}} = r \mid \mathbf{X}_1, \ldots, \mathbf{X}_K)$, $r = 3, 4, 5$. The posterior mode of R_{eff} equals the true value 3 in 98% of the replications in both cases.

case	block (k)	dim (r)	$\sigma_k^{(r)}$	$\hat{\sigma}_k^{(r)}$	Std. Dev	Dim r active on block k	Selected (%)
case=1	1	1	0.70	0.70	0.01	Yes	100.0
	1	2	0.50	0.50	0.01	Yes	100.0
	1	3	0.00	0.00	0.00	No	0.2
	1	4	0.00	0.00	0.00	No	0.0
	1	5	0.00	0.00	0.00	No	0.0
	2	1	0.60	0.60	0.01	Yes	100.0
	2	2	0.40	0.40	0.01	Yes	100.0
	2	3	0.00	0.00	0.00	No	1.2
	2	4	0.00	0.00	0.01	No	1.7
	2	5	0.00	0.00	0.00	No	0.0
	3	1	0.50	0.50	0.01	Yes	100.0
	3	2	0.00	0.00	0.00	No	0.4
	3	3	0.60	0.60	0.01	Yes	100.0
	3	4	0.00	0.00	0.00	No	0.0
	3	5	0.00	0.00	0.00	No	0.2
case=2	1	1	0.70	0.70	0.01	Yes	100.00
	1	2	0.10	0.10	0.01	Yes	99.70
	1	3	0.00	0.00	0.00	No	0.10
	1	4	0.00	0.00	0.00	No	0.20
	1	5	0.00	0.00	0.00	No	0.10
	2	1	0.60	0.60	0.01	Yes	100.00
	2	2	0.10	0.10	0.01	Yes	99.90
	2	3	0.00	0.00	0.00	No	1.50
	2	4	0.00	0.00	0.01	No	1.00
	2	5	0.00	0.00	0.00	No	0.00
	3	1	0.50	0.50	0.00	Yes	100.00
	3	2	0.00	0.00	0.00	No	0.10
	3	3	0.10	0.09	0.02	Yes	94.90
	3	4	0.00	0.00	0.02	No	4.70
	3	5	0.00	0.00	0.01	No	0.40

Table 1: Estimation of the block weights over $B = 1000$ replications: true value, mean and standard deviation of the posterior mean, and percentage of replications in which the weight is selected ($\hat{P}(\gamma_k^{(r)} = 1 \mid \mathbf{X}_1, \ldots, \mathbf{X}_K) > 1/2$).

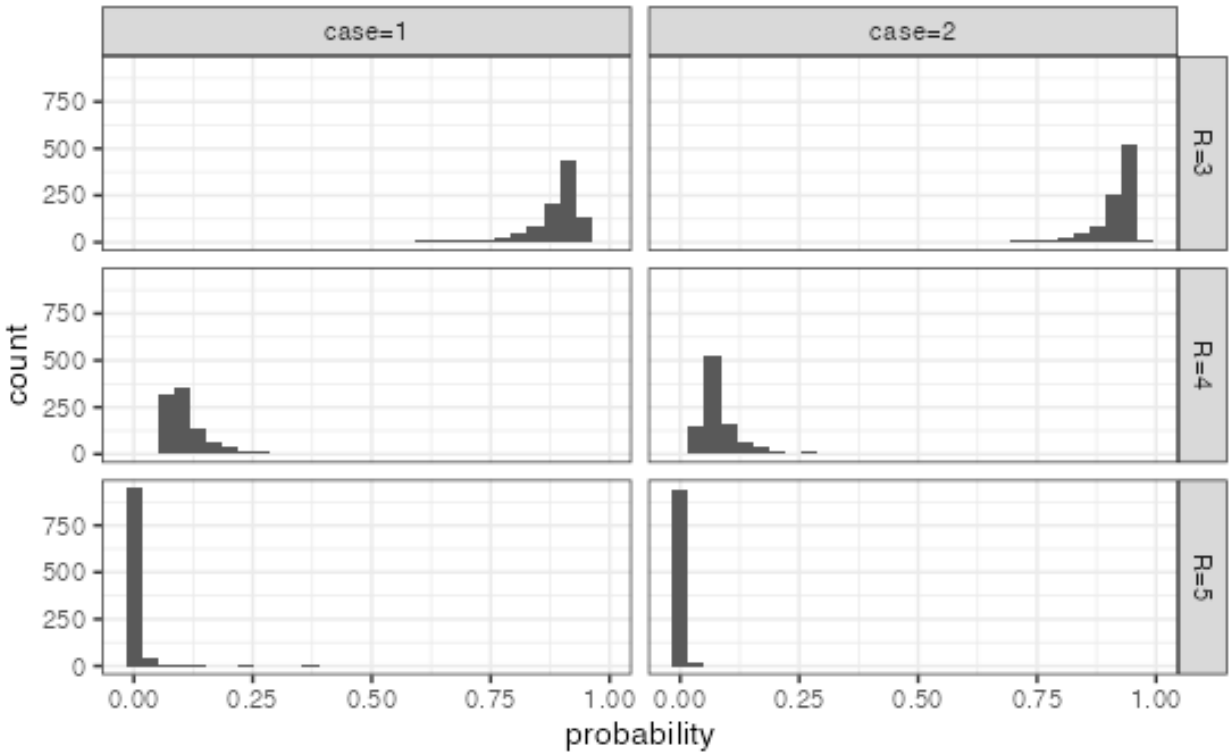


Figure 1: Distribution of the Bayes estimator of the probability of effective rank to be equal to 3, 4, 5 in both case.

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Appendix: proof of the proposition

Proposition 1. The criterion to be minimized is:

$$\mathcal{L}(\mathbf{U}, \mathbf{V}_1, \ldots, \mathbf{V}_K, \Sigma_1, \ldots, \Sigma_K) = \sum_{k=1}^{K} \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\top\|^2,$$

subject to $\mathbf{U}^\intercal\mathbf{U} = I_R$ and $\mathbf{V}_k^\intercal\mathbf{V}_k = I_R$ for all $k = 1, \ldots, K$.

The algorithm is of the *Alternating Least Squares* (ALS) type: at each iteration, the criterion is minimized successively with respect to each parameter, while the others are held fixed. It therefore suffices to show that each individual update decreases or maintains $\mathcal{L}$.

Update of U With $\mathbf{V}_k$ and Σ_k fixed, let $\mathbf{H}_k = \Sigma_k \mathbf{V}_k^\intercal \in \mathcal{M}_{R,p_k}(\mathbb{R})$ and $\mathbf{H} = [\mathbf{H}_1 \mid \cdots \mid \mathbf{H}_K] \in \mathcal{M}_{R,P}(\mathbb{R})$. Then:

$$\mathcal{L} = \sum_{k=1}^{K} \|\mathbf{X}_k - \mathbf{U}\mathbf{H}_k\|^2 = \|\mathbf{X} - \mathbf{U}\mathbf{H}^\intercal\|^2.$$

Expanding: $\|\mathbf{X} - \mathbf{U}\mathbf{H}^\intercal\|^2 = \|\mathbf{X}\|^2 - 2\ \mathrm{trace}(\mathbf{X}^\intercal\mathbf{U}\mathbf{H}^\intercal) + \|\mathbf{U}\mathbf{H}^\intercal\|^2$.

Since $\mathbf{U}^\intercal\mathbf{U} = I_R$: $\|\mathbf{U}\mathbf{H}^\intercal\|^2 = \mathrm{trace}(\mathbf{H}\mathbf{U}^\intercal\mathbf{U}\mathbf{H}^\intercal) = \mathrm{trace}(\mathbf{H}\mathbf{H}^\intercal) = \|\mathbf{H}\|^2$ (constant w.r.t. $\mathbf{U}$).

Minimizing $\mathcal{L}$ with respect to $\mathbf{U}$ is therefore equivalent to maximizing

$$\mathrm{trace}(\mathbf{X}^\intercal\mathbf{U}\mathbf{H}^\intercal) = \mathrm{trace}(\mathbf{H}^\intercal\mathbf{X}^\intercal\mathbf{U}) = \mathrm{trace}(\mathbf{U}^\intercal\mathbf{X}\mathbf{H}),$$

subject to $\mathbf{U}^\intercal\mathbf{U} = I_R$. The first equality comes from $\mathrm{trace}(\mathbf{AB}) = \mathrm{trace}(\mathbf{AB})$ for $\mathbf{A}, \mathbf{B}$ of dimension N, P and P, N. The second equality comes from the equality of the trace of a square matrix and its transpose. By [Cliff, 1966], the maximum is attained at $\mathbf{U}^* = AB^\intercal$, where A (resp. B) are the matrices of the R leading left (resp. right) singular vectors of $\mathbf{X}\mathbf{H}^T$. subject to $\mathbf{U}^\intercal\mathbf{U} = I_R$. The first equality comes from $\mathrm{trace}(\mathbf{AB}) = \mathrm{trace}(\mathbf{AB})$ for $\mathbf{A}, \mathbf{B}$ of dimension N, P and P, N. The second equality comes from the equality of the trace of a square matrix and its transpose. By [Cliff, 1966], the maximum is attained at $\mathbf{U}^* = AB^\intercal$, where A (resp. B) are the matrices of the R leading left (resp. right) singular vectors of $\mathbf{X}\mathbf{H}^T$. This update is the global minimizer of $\mathcal{L}$ with respect to $\mathbf{U}$, hence:

$$\mathcal{L}(\mathbf{U}^*, \ldots) \leq \mathcal{L}(\mathbf{U}_{\text{old}}, \ldots).$$

Update of $\mathbf{V}_k$ With $\mathbf{U}$ and Σ_k fixed, expanding the contribution of block k:

$$\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X}_k\|^2 - 2\ \mathrm{trace}(\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal) + \|\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2.$$

Since $\mathbf{U}^\intercal\mathbf{U} = I_R$ and $\mathbf{V}_k^\intercal\mathbf{V}_k = I_R$:

$$\|\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \mathrm{trace}(\mathbf{V}_k\Sigma_k^\intercal\mathbf{U}^\intercal\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal) = \mathrm{trace}(\Sigma_k^2) \quad \text{(constant w.r.t. } \mathbf{V}_k\text{)}.$$

Minimizing therefore reduces to maximizing $\mathrm{trace}(\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal) = \mathrm{trace}(\mathbf{V}_k^\intercal\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k)$ subject to $\mathbf{V}_k^\intercal\mathbf{V}_k = I_R$. Since the diagonal matrix $\Sigma_k > 0$ does not affect the singular vectors, the maximum is attained at $\mathbf{V}_k^* = \mathbf{A}\mathbf{B}^\intercal$ where $\mathbf{A}$, $\mathbf{B}$ are obtained from the SVD of $\mathbf{X}_k^\intercal\mathbf{U}$ [Cliff, 1966] . Hence: subject to $\mathbf{V}_k^\intercal\mathbf{V}_k = I_R$. Since the diagonal matrix $\Sigma_k > 0$ does not affect the singular vectors, the maximum is attained at $\mathbf{V}_k^* = \mathbf{A}\mathbf{B}^\intercal$ where $\mathbf{A}$, $\mathbf{B}$ are obtained from the SVD of $\mathbf{X}_k^\intercal\mathbf{U}$ [Cliff, 1966] . Hence:

$$\mathcal{L}(\ldots, \mathbf{V}_k^*, \ldots) \leq \mathcal{L}(\ldots, \mathbf{V}_{k,\text{old}}, \ldots).$$

Update of Σ_k With $\mathbf{U}$ and $\mathbf{V}_k$ fixed (both having orthonormal columns), and $\Sigma_k = \mathrm{diag}(\sigma_k^{(1)}, \ldots, \sigma_k^{(R)})$, the contribution of block k is:

$$\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X}_k\|^2 - 2\ \mathrm{trace}(\mathbf{V}_k^\intercal\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k) + \mathrm{trace}(\Sigma_k^2).$$

Since Σ_k is diagonal, it remains to minimazing:

$$\sum_{r=1}^{R}\Big[()\sigma_k^{(r)})^2 - 2\,(\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)_{rr} \cdot d_r\Big].$$

Each term is a convex parabola in $\sigma_k^{(r)}$, minimized at: $(\sigma_k^{(r)})^* = (\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)_{rr}$, or, in matrix form:$\Sigma_k^* = \mathrm{diag}(\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)$. This update is therefore the exact minimizer with respect to Σ_k, and:

$$\mathcal{L}(\ldots, \Sigma_k^*, \ldots) \leq \mathcal{L}(\ldots, \Sigma_{k,\text{old}}, \ldots).$$

Global Convergence At each complete iteration (update of $\mathbf{U}$, then all $\mathbf{V}_k$, then all Σ_k), the criterion $\mathcal{L}$ is monotonically decreasing and bounded below by zero. By the monotone convergence theorem, the sequence of values $(\mathcal{L}^{(t)})_{t\geq 0}$ converges:

$$\mathcal{L}^{(t+1)} \leq \mathcal{L}^{(t)}, \quad \mathcal{L}^{(t)} \geq 0 \implies \mathcal{L}^{(t)} \xrightarrow[t\to\infty]{} \ell^* \geq 0.$$

□

Proposition 2. (1)–(2): by (4), the conditional density of $\mathbf{U}$ is proportional to $\exp\{\sum_k \tau_k \operatorname{trace}(\Sigma_k \mathbf{V}_k^\intercal \mathbf{X}_k^\intercal \mathbf{U})\} = \exp\{\operatorname{trace}(\mathbf{F}_\mathbf{U}^\intercal \mathbf{U})\}$, and $\operatorname{trace}(\mathbf{F}_\mathbf{U}^\intercal \mathbf{Q}\tilde{\mathbf{U}}) = \operatorname{trace}\big((\mathbf{Q}^\intercal \mathbf{F}_\mathbf{U})^\intercal \tilde{\mathbf{U}}\big)$; the same argument applies to $\mathbf{V}_k$ with $\operatorname{trace}(\Sigma_k \mathbf{V}_k^\intercal \mathbf{X}_k^\intercal \mathbf{U}) = \operatorname{trace}(\mathbf{F}_{\mathbf{V}_k}^\intercal \mathbf{V}_k)/\tau_k$. (3): the terms of (4) involving $\sigma_k^{(h)}$ are $-\frac{\tau_k}{2}[(\sigma_k^{(h)})^2 - 2\sigma_k^{(h)} d_k^{(h)}]$; adding the log-prior $-\frac{\lambda}{2}(\sigma_k^{(h)})^2$ and completing the square gives a Gaussian kernel, restricted to $\sigma_k^{(h)} > 0$. (4): the Gamma prior is conjugate for the precision of a Gaussian likelihood. □

Proposition 3. Items 1, 2, 4 and 5 follow as in the non-sparse sampler (linearity of (4) in $\mathbf{U}$ and $\mathbf{V}_k$, and conjugacy). For item 3, the terms of the log-posterior involving $\sigma_k^{(r)}$ are, by (4), $\tau_k d_k^{(r)} \sigma_k^{(r)} - \frac{\tau_k}{2}(\sigma_k^{(r)})^2$. Under $\gamma_k^{(r)} = 0$, $\sigma_k^{(r)} = 0$ and this quantity is 0. Under $\gamma_k^{(r)} = 1$, integrating against the half-normal slab of density $2\sqrt{\lambda/(2\pi)}\exp(-\lambda s^2/2)$ on $(0,\infty)$ gives

$$2\sqrt{\frac{\lambda}{2\pi}} \int_0^\infty \exp\Big(-\frac{\omega_k}{2} s^2 + \tau_k d_k^{(r)} s\Big) ds = 2\sqrt{\frac{\lambda}{\omega_k}} \exp\Big(\frac{\omega_k (m_k^{(r)})^2}{2}\Big) \Phi\big(m_k^{(r)} \sqrt{\omega_k}\big) = B_k^{(r)},$$

which is the Bayes factor of $\gamma_k^{(r)} = 1$ against $\gamma_k^{(r)} = 0$. The conditional of $\sigma_k^{(r)}$ given $\gamma_k^{(r)} = 1$ is obtained by completing the square. □